

Network Settings and Commands on Windows

Computer Networks 1 (Lab) · Session 1

Session Objectives

1. Configure the network adapter on Windows.
2. Get to know the Windows CMD tools.

Skills You Will Gain

- Understand how the basic Windows network commands work.
- Use `ipconfig`, `ping`, `netstat` and `tracert` to test connectivity and analyze the network.
- Configure the Network Adapter and its settings.
- Analyze command output and draw conclusions about connection status.

Required Tools

- A computer running Windows.
- Administrator privileges to run the Command Line Interface (CMD).

1. Background

A **network** is a group of computing devices connected together to exchange data and share resources. Each device connects through a **network adapter** (network card), which can be wired (Ethernet) or wireless (Wi-Fi). The adapter holds the settings that identify the device on the

network, such as its IP address, subnet mask, default gateway and DNS server.

Protocol: the set of rules that governs how devices communicate with each other.

IP address: a logical numeric address given to each device so it can be identified and reached, for example 192.168.1.10 .

In this session you will view these settings, change them, and use CMD commands to check that the network works.

2. Setting Up the Network Adapter

Step 1: Prepare your work environment

- Make sure the computer is connected to a local network and to the Internet.
- Open CMD and choose **Run as Administrator**.

Step 2: Open the adapter settings

Control Panel → Network and Internet → Network and Sharing Center

Select the adapter you want (Ethernet or Wi-Fi) and inspect its properties.

Step 3: Set the IP address manually

1. Open Settings → Network & Internet, then select your current connection (Ethernet or Wi-Fi).
2. Click **Properties** and open the **IP settings**.
3. Change the mode from automatic (**DHCP**) to **Manual**.
4. Enter the network information, then save the changes.

Field	Description	Example
IP address	The address you want for the device	192.168.1.10
Subnet mask	Usually 24 bits	255.255.255.0
Default gateway	The router's address	192.168.1.1
Preferred DNS server	Translates names to IP addresses	8.8.8.8

3. CMD Commands

Command	What it does	Example
ipconfig	Shows the device's IP address, subnet mask and default gateway. Add /all for more details (such as MAC address and DNS).	ipconfig /all
ping	Tests the connection between your device and another one, such as a server.	ping 8.8.8.8 ping www.google.com
netstat	Shows active connections and open ports. With -an it lists all ports and their states (LISTENING / ESTABLISHED).	netstat -an
tracert	Shows the path packets take to reach the destination (a server, for example).	tracert www.google.com

How to read the results

- **ping:** a "Reply" means the connection works; "Request timed out" means no response; `time=` is the response time.
- **tracert:** each line is one router (hop) on the path.
- **netstat:** note which ports are listening and which connections are established.

4. Practical Exercise

1. Run the following in CMD:

- Show the network settings of your computer.
- Test connectivity with the Yarmouk University server `ypu.edu.sy`.
- Show the path the packets took in the previous test.
- Show the active ports on your computer.

2. Save the results in a file named `task1.txt`.

Tip: you can send a command's output to a file with `>` (overwrite) or `>>` (append), for example `ipconfig /all > task1.txt`.

End of session